How many Observations are Enough? Knowledge Distillation for Trajectory Forecasting

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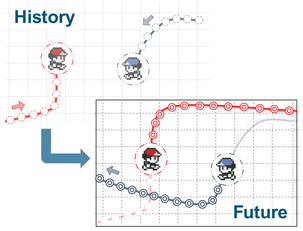
1. Pedestrian Trajectory Forecasting

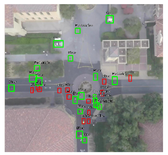
SOTA models [1, 2, 4, 3] rely on a **history** of past tracked locations (8 time-steps ≈ 3 to 5 seconds).

GOAL: predicting a plausible sequence of **future locations** (12 time-steps ≈ up to the next 5 sec.).

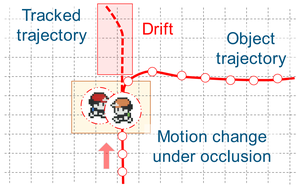


2. Main Problems

In realistic applications, the collection of input trajectories involves **machine perception** (i.e., detection and tracking).



In crowded scenes, detection and fragmentation errors may accumulate, leading to **tracking drifts**.



Implications: the *longer* the input sequences, the *higher* the chance the model will be fed with corrupted and noisy input data, thus fatally **affecting its prediction performance**.

3. Our Idea

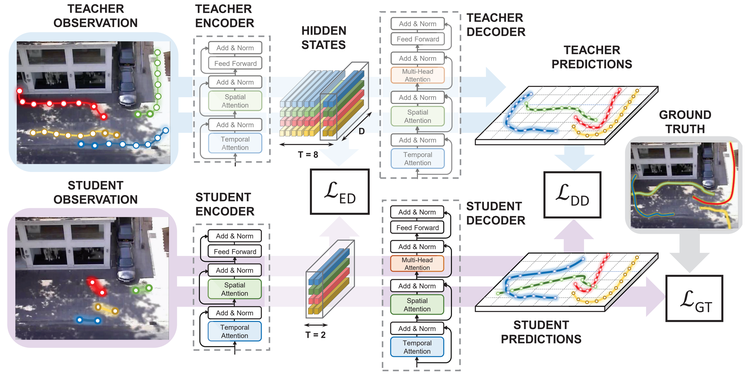
IDEA: reducing the length of the input trajectories to the extent we can minimize the risks of automatic perception.

The *shorter* they are:
⇒ less error-prone ☺
⇒ less informative ☹

Distilling the Observations (DTO): a strategy to obtain **accurate predictions** with only few input observations (*e.g.*, just two ones).

4. A Teacher-Student Approach

DTO builds upon (self-)Knowledge Distillation: the student has to mimic its teacher by leveraging fewer observations, i.e., the two last observations only.



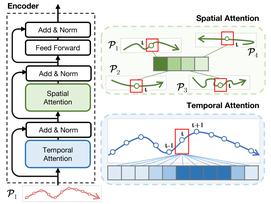
Learning by filling INFORMATION GAPS

First, we train a teacher network given standard 8-length observation sequences.

Second, we freeze its parameters and attempt to **transfer its predictive capability to a student network** (featuring the same architectural design).

5. Spatio-Temporal Transformer

STT: a transformer-like architecture modelling temporal and spatial cues through two modules.



A **temporal encoder** leverages **temporal patterns** between following time steps.

A **spatial encoder** exploits **human-human interactions** between agents at a fixed time.

6. Comparison With The SOTA

	ETH/UCY	SDD	Lyft
Ind-TF [1]	0.50 / 0.96	0.74 / 1.46	0.31 / 0.62
CVM [2]	0.53 / 1.15	0.69 / 1.39	0.29 / 0.61
SR-LSTM [4]	0.45 / 0.94	0.72 / 1.47	**0.20** / **0.43**
STAR [3]	**0.41** / **0.87**	-	-
STT (8 obs)	0.43 / 0.88	**0.63** / **1.26**	0.24 / 0.53
STT (2 obs)	0.57 / 1.12	0.73 / 1.44	0.31 / 0.56
DTO (2 obs)	0.46 / 0.93	0.64 / 1.27	0.27 / 0.55

DtO boosts the student, approaching its teacher (STT 8 obs) and SOTA results.

7. On Tracked Trajectories (SDD)

Scene	STT (8 obs)	STT (2 obs)	CVM (2 obs)	DTO (2 obs)
bookstore	0.58 / 1.08	0.54 / 1.01	0.55 / 1.02	**0.53** / **0.99**
nexus	1.38 / 2.10	1.33 / 2.07	1.38 / 2.15	**1.29** / **2.03**
deathCircle	0.99 / 1.83	0.97 / 1.83	0.99 / 1.86	**0.94** / **1.82**
gates	1.20 / 2.15	1.00 / 1.94	1.10 / 1.97	**0.94** / **1.84**
hyang	**0.39** / **0.82**	0.48 / 0.95	0.41 / 0.85	0.46 / 0.89
coupa	0.28 / 0.49	0.21 / 0.41	0.26 / 0.44	**0.20** / **0.38**
overall	0.84 / 1.46	0.80 / 1.40	0.84 / 1.40	**0.77** / **1.37**

With a tracker in the middle, DTO provides more robustness w.r.t. the teacher.

References

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[3] C. Yu et al. Spatio-temporal graph transformer networks for pedestrian trajectory prediction. In *ECCV*, 2020.
[4] P. Zhang et al. Sr-lstm: State refinement for lstm towards pedestrian trajectory prediction. In *CVPR*, 2019.